

Stage 4A — Multi-Neighbour Model Selection Report

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1 Introduction

This stage report documents the Stage 4A multi-neighbour simulation-based model selection for synchronisation algorithms prior to hardware-in-the-loop (HIL) testing. Five candidate models were evaluated under identical decentralised multi-agent conditions using a pure-software simulator. The goal is to select a primary and a secondary HIL candidate based on strict zero-lag synchronisation performance, robustness across topologies and frequency sets, N=5 scalability, hardware readiness, and interpretability.

2 Candidate Models

All models produce flash outputs and are evaluated with common flash-based synchronisation metrics. The PCO and EAPF variants are event-driven: each agent updates its state only when neighbour flash events are detected. Kuramoto uses continuous phase coupling in simulation and would require phase reconstruction from flash timestamps for hardware implementation. Locked parameters were selected via a $60\text{ s} \times 5\text{-repeat}$ parameter sweep (seeds 1000–1004) across four frequency sets and three topologies.

Table 1: Locked model parameters selected via $60\text{ s} \times 5\text{-repeat}$ parameter-locking sweep (seeds 1000–1004).

Model	Key Parameters	Variant
Kuramoto	$K = 5.0$	baseline
PCO simple	additive phase, $\varepsilon = 0.10$, $\tau_{\text{ref}} = 0.05\text{ s}$	simple
PCO adaptive PRC	biphasic sine PRC, $\varepsilon = 0.10$, phase delay on, $\Delta\phi_{\text{max}} = 0.20$	adaptive_prc
EAPF tracker	$g_p = 0.40$, $g_f = 0.15$	tracker
EAPF consensus	$g_p = 0.02$, $g_f = 0.02$, $\alpha_{p,f} = 0.2$, $\Delta\theta_{\text{max}} = 0.2\text{ rad}$, $\Delta f_{\text{max}} = 0.05\text{ Hz}$	consensus

Table 2: Final weighted model evaluation scores (N=3: 1200 trials, N=5: 300 trials, seeds 2000–2019 / 3000–3009).

Model	N3 Sync	N3 Conv	N3 Robust	N5 Scale	HW	Int.	Total
EAPF consensus	1.000	0.984	1.000	0.933	0.800	0.800	0.938
Kuramoto	0.902	0.891	0.767	0.427	0.850	0.900	0.801
EAPF tracker	0.763	0.752	0.335	0.597	0.700	0.700	0.656
PCO adaptive	0.350	0.467	0.100	0.000	0.650	0.750	0.363
PCO simple	0.283	0.370	0.100	0.000	0.750	0.900	0.358

Table 3: Repeatability across two independent final evaluations (seeds 2000/3000 vs. 4000/5000).

Model	Run 1	Run 2	Δ
EAPF consensus	0.9375	0.9375	0.0000
Kuramoto	0.8008	0.8020	+0.0012
EAPF tracker	0.6563	0.6602	+0.0039
PCO adaptive	0.3625	0.3625	0.0000
PCO simple	0.3579	0.3579	0.0000

Table 4: Final Stage 4A model-selection decision summary.

Rank	Model	Decision
1	EAPF consensus	Primary HIL candidate
2	Kuramoto	Secondary comparison model
3	EAPF tracker	Retained — works for near-identical oscillators
4	PCO adaptive PRC	Retained — reduced extra flashes vs. simple PCO
5	PCO simple	Retained — simplest firefly-inspired baseline

2.1 Kuramoto (continuous phase coupling)

Working principle [?]. Each agent i continuously observes the phase of its visible neighbours and adjusts its own phase velocity according to the Kuramoto ODE:

$$\dot{\theta}_i = \omega_i + \frac{K}{N_i} \sum_{j \in \mathcal{N}_i} \sin(\theta_j - \theta_i) \quad (1)$$

where $\mathcal{N}_i$ is the neighbour set of agent i , $N_i = |\mathcal{N}_i|$, and K is the coupling strength.

Locked parameter: $K = 5.0$.

Strengths: mathematically well-understood; achieves 91.7% zero-lag success in N=3; simple to implement on hardware using phase reconstruction from flash timestamps.

Limitations: requires continuous phase observability which the camera pipeline approximates via flash-to-phase estimation; N=5 scalability limited (33.3% zero-lag success).

2.2 PCO Simple (event-driven pulse coupling)

Working principle [?, ?]. Each agent maintains a normalised phase $\phi \in [0, 1)$. On detecting a neighbour flash, the phase is additively advanced:

$$\phi \leftarrow \min(1.0, \phi + \varepsilon) \quad (2)$$

When $\phi \geq 1.0$ the agent fires and resets to 0, entering a refractory period. This is the simplest firefly-inspired pulse-coupled oscillator, derived from the Mirollo–Strogatz model.

Locked parameters: additive phase, $\varepsilon = 0.10$, $\tau_{\text{ref}} = 0.05$ s.

Strengths: directly event-driven; matches the camera detection pipeline; conceptually simplest firefly model.

Limitations: 25.0% zero-lag success at N=3; 0% at N=5; flash count ratio 1.308 (extra flashes in heterogeneous conditions).

2.3 PCO Adaptive PRC (phase response curve variant)

Working principle [?, ?]. Extends PCO simple with a biphasic sine phase response curve:

$$\Delta\phi = \varepsilon \cdot \sin(2\pi(\phi - \phi_{\text{ref}})) \quad (3)$$

Phase delay is enabled (negative corrections allowed), and maximum correction is rate-limited. Optional frequency adaptation is available but was not used in the locked variant.

Locked parameters: biphasic sine PRC, $\varepsilon = 0.10$, phase delay enabled, $\Delta\phi_{\text{max}} = 0.20$.

Strengths: lower flash count ratio (1.175 vs. 1.308 for simple PCO); better 1:1 flash lock rate (75% vs. 42%).

Limitations: 25.0% zero-lag success at N=3; 0% at N=5; still limited by excitatory pulse coupling under strong heterogeneity.

2.4 EAPF Tracker (fixed-leader phase/frequency tracker)

Working principle [?]. On each detected leader flash, the agent computes the signed phase error and applies independent phase and frequency corrections:

$$\theta \leftarrow \text{wrap}_{2\pi}(\theta + g_p \cdot e_\theta) \quad (4)$$

$$f \leftarrow \text{clamp}(f + g_f \cdot e_\theta / (2\pi), f_{\text{min}}, f_{\text{max}}) \quad (5)$$

where $e_\theta = \text{wrap}_\pi(0 - \theta)$.

Locked parameters: $g_p = 0.40$, $g_f = 0.15$.

Strengths: simple engineering tracker; 76.2% zero-lag success at N=3; works well for near-identical oscillators.

Limitations: designed for fixed-leader tracking; less robust in mutual multi-neighbour mode (flash count ratio 1.116).

2.5 EAPF Consensus (event-based phase/frequency consensus)

Working principle [?, ?]. Each agent maintains per-neighbour state estimates (period, frequency, phase) updated *only* from flash timestamps. Consensus corrections use the mean phase and frequency errors across all visible neighbours:

$$e_\theta = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \text{wrap}_\pi(\hat{\theta}_j - \theta_i) \quad (6)$$

$$e_f = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} (\hat{f}_j - f_i) \quad (7)$$

where $\hat{\theta}_j$ and $\hat{f}_j$ are the estimated neighbour phase and frequency. Corrections are low-pass filtered and rate-limited.

Locked parameters: $g_p = 0.02$, $g_f = 0.02$, $\alpha_p = \alpha_f = 0.2$, $\Delta\theta_{\text{max}} = 0.2 \text{ rad}$, $\Delta f_{\text{max}} = 0.05 \text{ Hz}$.

Strengths: 100% zero-lag success at N=3 *and* N=5; flash count ratio 1.000; nearest-flash timing error 6.6 ms; frequency spread 0.0 Hz; only model achieving perfect strict synchronisation across all conditions.

Limitations: very low gains require 60s for full convergence; more stateful than simpler models (per-neighbour estimates).

3 Evaluation

3.1 Evaluation Setup

The multi-agent simulator runs N agents at fixed time-step $dt = 0.01 \text{ s}$. Each agent observes neighbour flash events according to a graph-based topology.

N=3 main evaluation. The primary HIL-aligned evaluation uses 3 agents with topologies `all_to_all`, `chain`, and `directed_chain`, and four frequency sets: identical [2.0, 2.0, 2.0] Hz, near-identical [1.9, 2.0, 2.1] Hz, moderate heterogeneity [1.8, 2.0, 2.2] Hz, and strong heterogeneity [1.5, 2.0, 2.3] Hz. For each model, this gives $4 \times 3 \times 20 = 240$ trials. Across all five models, the N=3 evaluation comprises $5 \times 4 \times 3 \times 20 = 1200$ trials, each of 60 s duration with independent random seeds (2000–2019).

N=5 scalability evaluation. A second evaluation with 5 agents uses three local topologies (`local_ring_5`, `chain_5`, `local_degree_2_3`) and two frequency sets (near-identical and strong heterogeneity), 10 repeats per combination (seeds 3000–3009), producing $5 \times 3 \times 2 \times 10 = 300$ trials total.

Repeatability. A second independent run with seeds 4000–4019 (N=3) and 5000–5009 (N=5) was performed with all locked parameters unchanged.

3.2 Metrics

Synchronisation quality. Zero-lag group sync requires phase synchronisation (pairwise timing error < 0.10 s and order parameter $R > 0.85$), frequency lock (spread < 0.05 Hz), and 1:1 flash lock (flash count ratio ≤ 1.2). Phase-locked group sync relaxes the in-phase constraint and allows stable non-zero offsets (offset jitter < 0.05 s).

Convergence and steady-state. Time-to-sync is the first time the 5-cycle sustained criterion is met. Steady-state metrics (final 10 s) include mean pairwise timing error, offset jitter, frequency spread, and order parameter.

Flash behaviour. Flash count ratio is the maximum final-window flash count divided by the minimum, measuring extra-flash or harmonic-locking behaviour.

3.3 Weighted Evaluation Matrix

The final model score is a weighted sum of six categories: N=3 synchronisation performance (0.30), N=3 convergence/quality (0.15), N=3 robustness across conditions (0.15), N=5 scalability (0.15), hardware readiness (0.15, expert-assigned), and interpretability (0.10, expert-assigned).

4 Results and Discussion

4.1 N=3 Performance

EAPF consensus achieved 100% zero-lag group sync across all 240 N=3 trials for that model (4 frequency sets $\times$ 3 topologies $\times$ 20 repeats). Kuramoto achieved 91.7% zero-lag success, with failures concentrated in chain and directed_chain topologies under strong heterogeneity. EAPF tracker achieved 76.2%, and both PCO variants achieved 25.0%.

The flash count ratio (Table ??) clearly separates models: EAPF consensus and Kuramoto maintain FCR = 1.0; EAPF tracker (1.12) and PCO adaptive PRC (1.18) show moderate extra flashes; PCO simple (1.31) shows significant extra flashes indicating repeated near-simultaneous group flash clusters with additional isolated flashes from faster agents.

4.2 N=5 Scalability

N=5 sharply separates the models. EAPF consensus is the only model maintaining 100% zero-lag sync. EAPF tracker drops to 58.3%, Kuramoto to 33.3%, and both PCO variants to 0%. The N=5 results confirm that consensus PLL with low gains scales to larger agent counts under local-neighbourhood topologies, while the other models degrade.

4.3 Model Ranking

Table ?? presents the final weighted scores. EAPF consensus ranks first (0.938), followed by Kuramoto (0.801), EAPF tracker (0.656–0.660), PCO adaptive PRC (0.363), and PCO simple (0.358).

4.4 Repeatability

The second independent evaluation (seeds 4000/5000) produced nearly identical scores (Table ??): the maximum delta was 0.0039 (EAPF tracker, +0.0039). All models are within the “very stable” band ($|\Delta| < 0.03$). The ranking is unchanged: EAPF consensus remains first, Kuramoto second.

4.5 Interpretation

Why EAPF consensus outperforms Kuramoto. EAPF consensus is a fully event-driven distributed frequency/phase agreement protocol [?, ?]. It converges slowly (low gains) but robustly, without overshoot or correction-induced extra flashes. At 60 s, even the 1.5–2.3 Hz 0.8 Hz frequency spread is resolved to 0.0 Hz final spread. Kuramoto’s continuous coupling is stronger but less robust under directed/sparse topologies.

Why Kuramoto remains strong but less scalable. Kuramoto [?] achieves 91.7% zero-lag at N=3 but drops to 33.3% at N=5 because continuous coupling weakens as the number of indirect neighbours increases and the effective coupling per link becomes diluted.

Why PCO variants remain limited. Excitatory phase-advance coupling [?] cannot decelerate an oscillator that is already too fast. Under strong heterogeneity, the 2.3 Hz agent consistently emits extra flashes. The adaptive PRC (biphasic sine with phase delay) reduces but does not eliminate this behaviour. This is consistent with the firefly-inspired synchronisation limitations discussed in [?].

4.6 Flash Raster Interpretation

Figure ?? shows representative flash rasters over the full 0–60 s interval for the all_to_all, strong heterogeneity condition. Figure ?? provides a 0–10 s zoom to reveal early-time convergence dynamics.

EAPF consensus (panel b) shows agents converging to a single synchronised flash cluster by approximately $t = 20$ s, with perfect 1:1 locking thereafter. Kuramoto (panel a) shows rapid convergence within the first 3–5 s. EAPF tracker (panel c) shows slower convergence with some residual offset. PCO adaptive PRC (panel d) and PCO simple (panel e) show clustered but non-1:1 flash patterns, with the fastest agent consistently emitting additional flashes between group events.

Raster plots should be interpreted together with quantitative metrics: the 0–60 s raster shows overall structure; early-time offsets may be visually compressed; the 0–10 s zoom reveals initial convergence; time-to-sync and timing-error metrics provide quantitative confirmation.

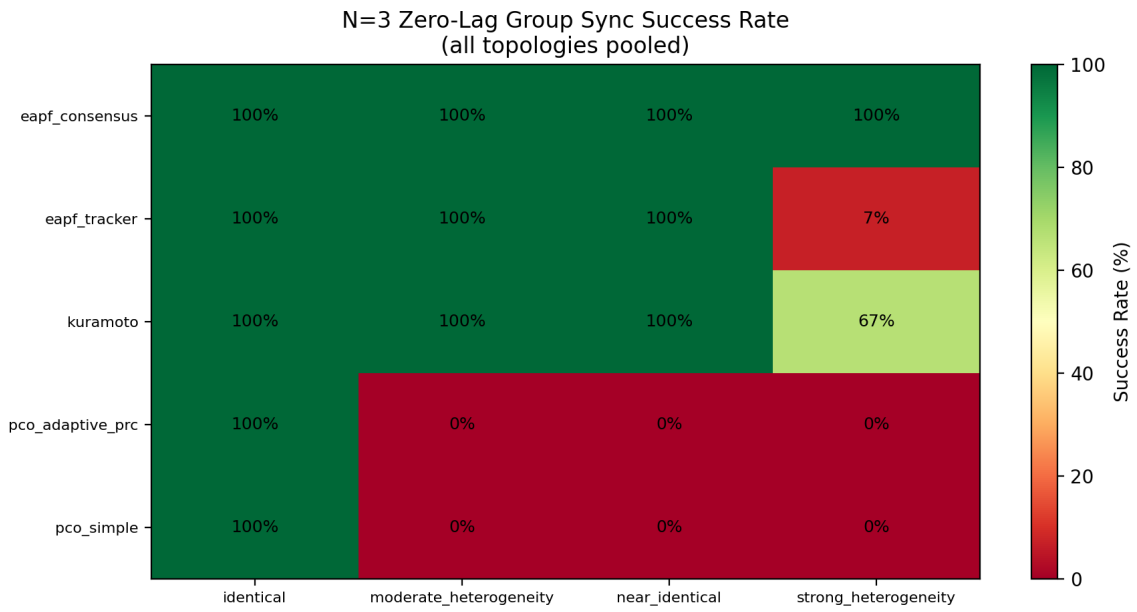


Figure 1: N=3 zero-lag group sync success rate by model and frequency set (all topologies pooled).

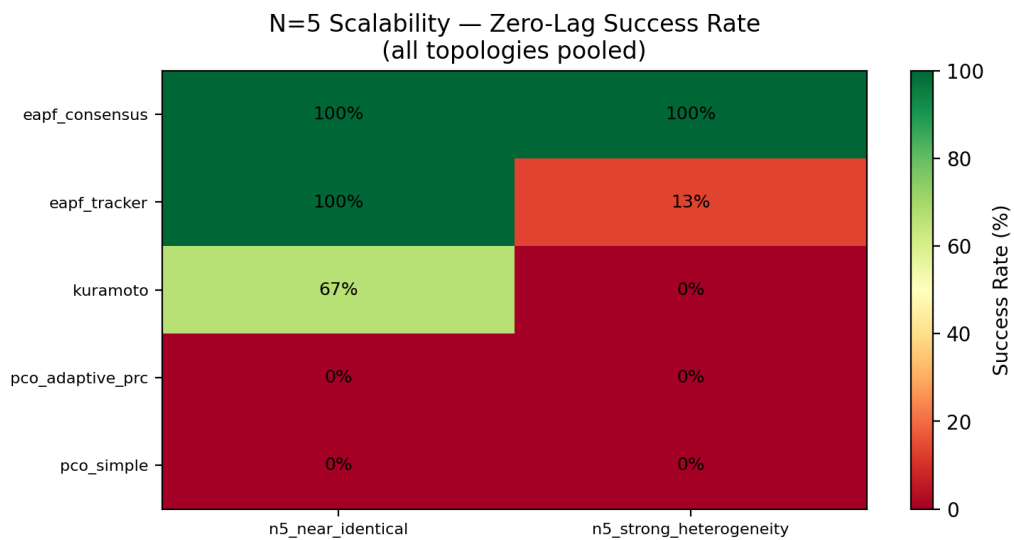


Figure 2: N=5 scalability — zero-lag success rate by model and frequency set (all topologies pooled).

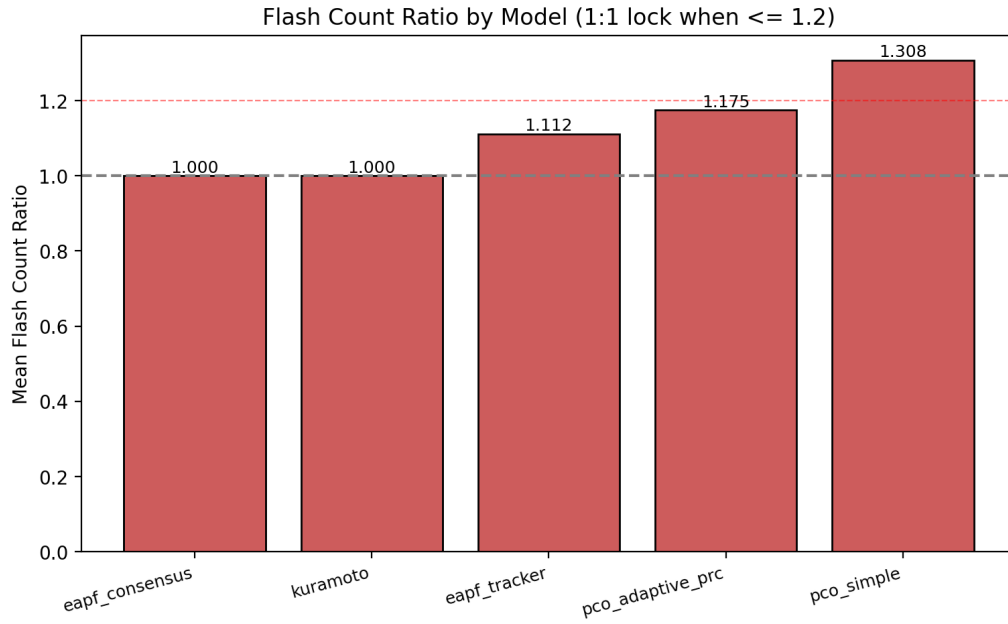


Figure 3: N=3 mean flash count ratio by model. The dashed line at 1.2 marks the 1:1 flash lock threshold.

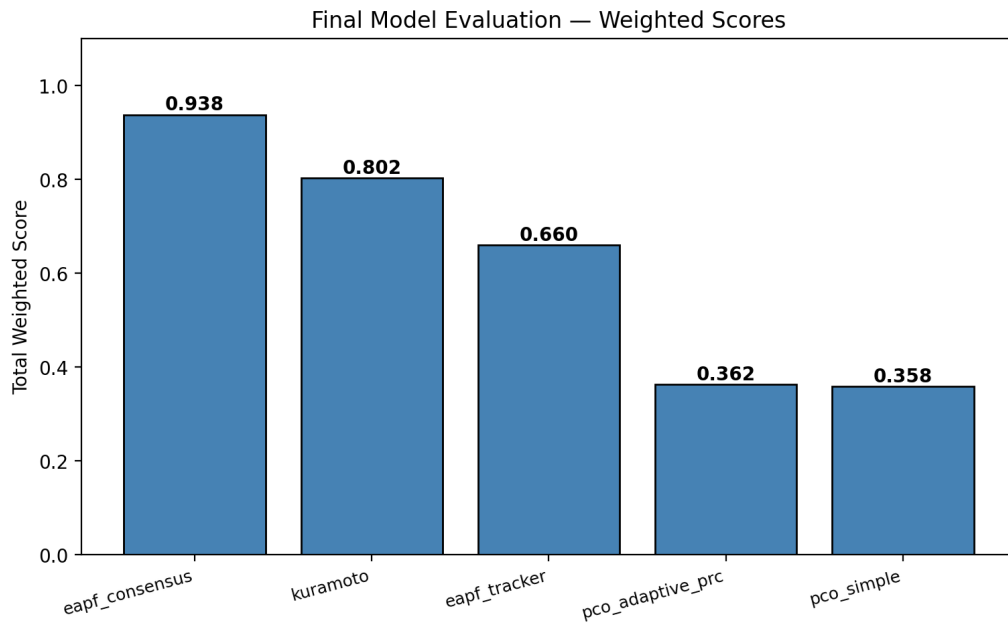


Figure 4: Total weighted model evaluation scores.

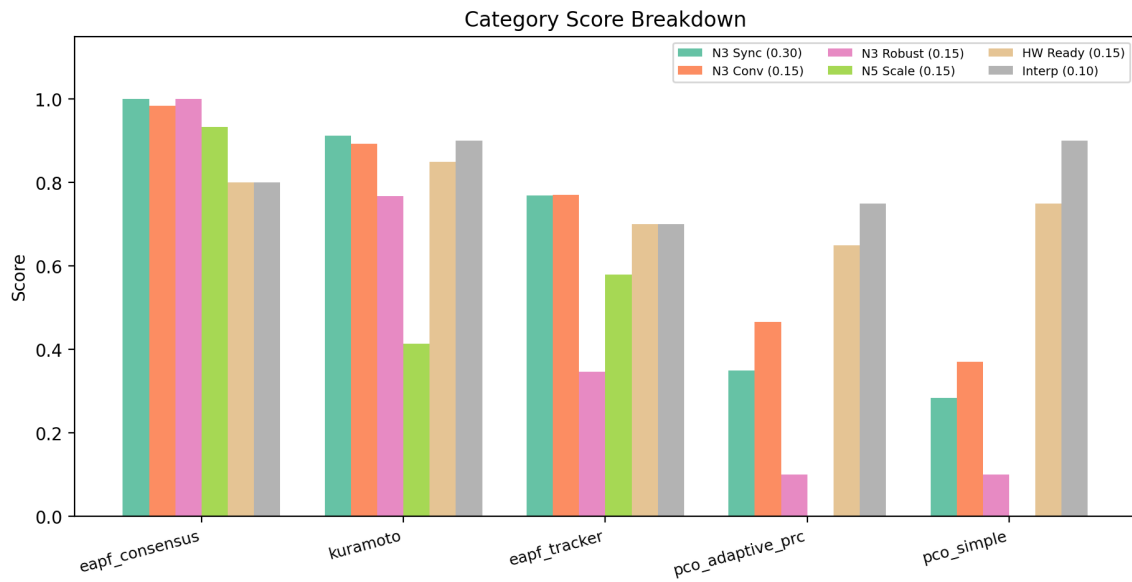


Figure 5: Category score breakdown: N3 Sync (0.30), N3 Conv (0.15), N3 Robust (0.15), N5 Scale (0.15), HW Ready (0.15), Interp (0.10).

Representative Flash Rasters — all_to_all, strong heterogeneity, 0-60 s

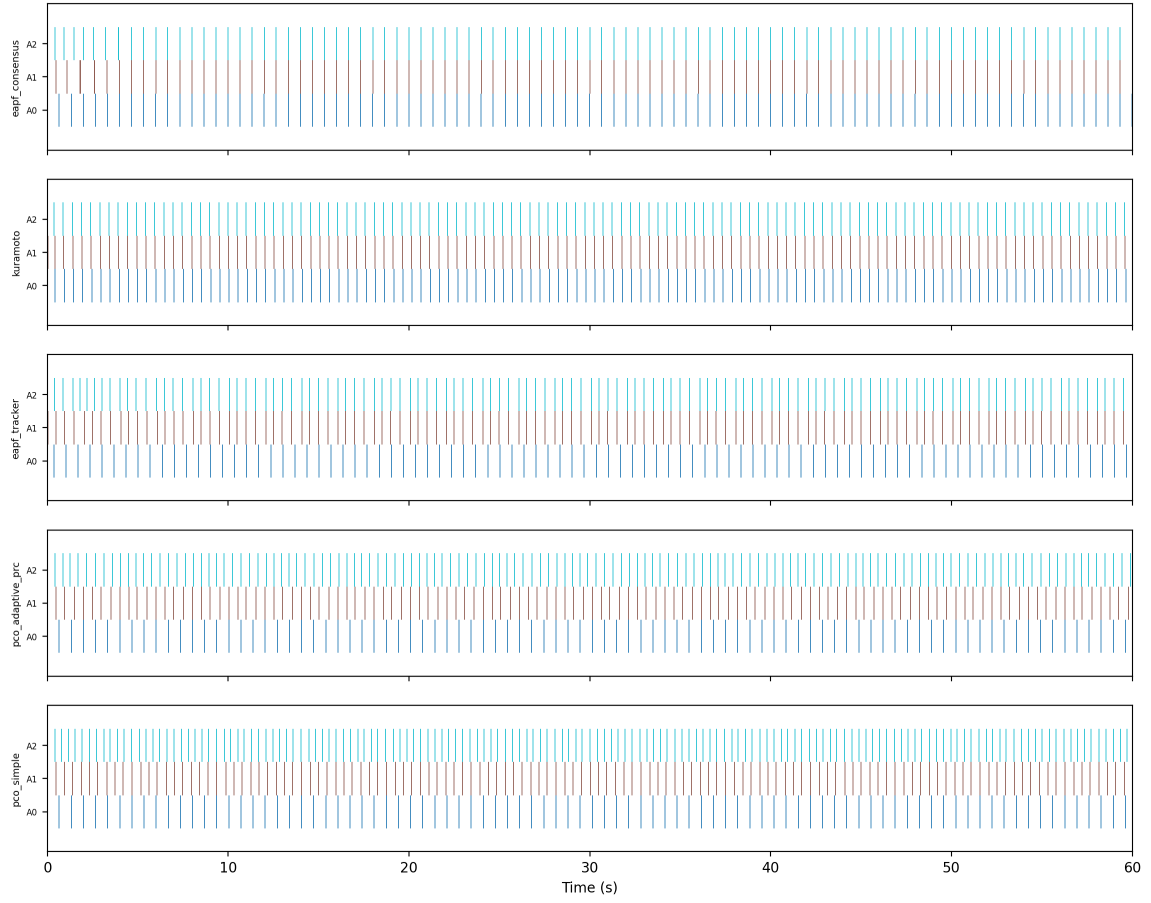


Figure 6: Representative flash rasters for all five models (all_to_all, strong heterogeneity, full 0–60s). Each panel shows agent flash events as vertical tick marks. (a) Kuramoto, (b) EAPF consensus, (c) EAPF tracker, (d) PCO adaptive PRC, (e) PCO simple. Extra flashes from faster agents are visible in panels (d) and (e).

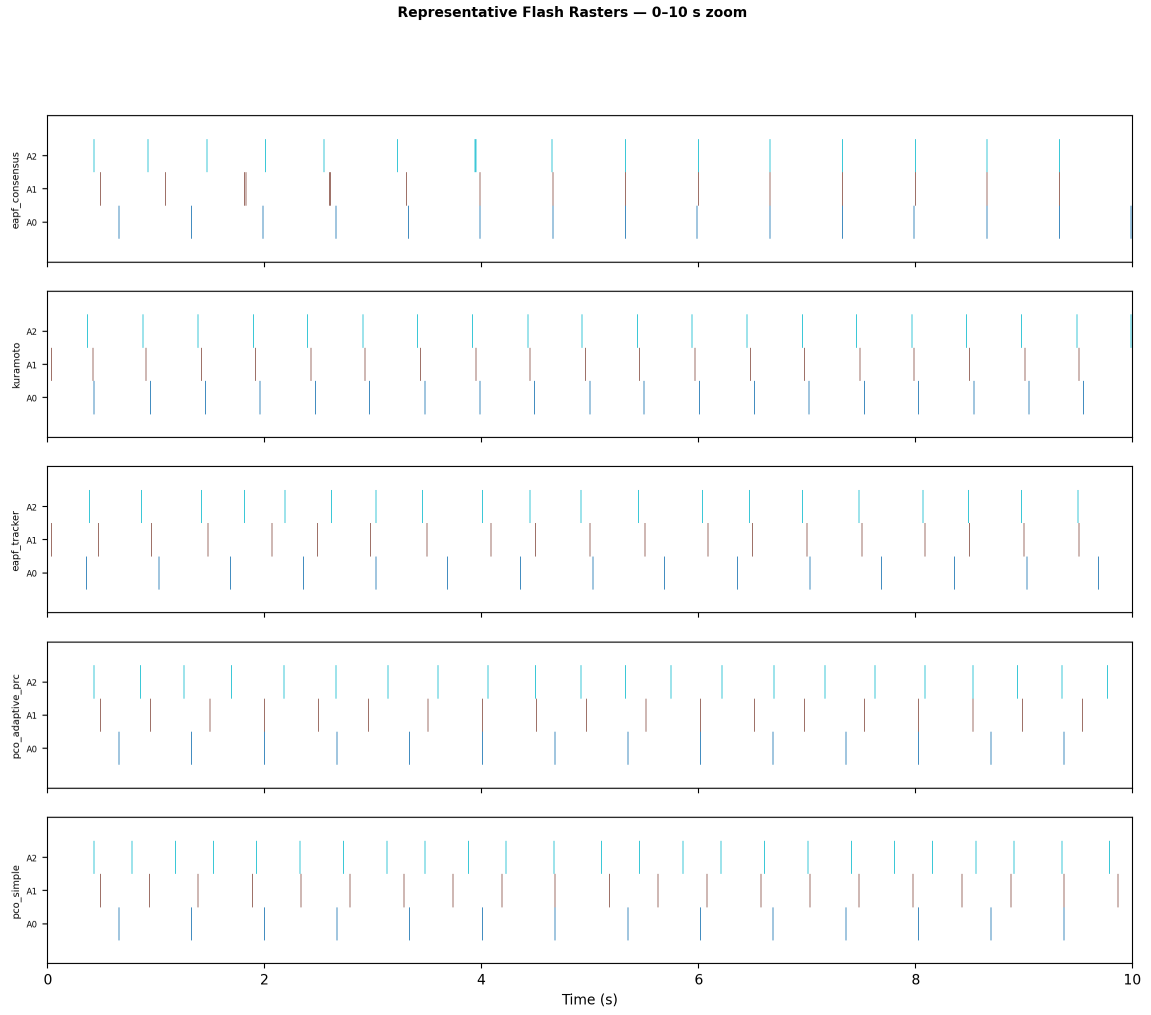


Figure 7: Representative flash rasters — 0–10s zoom showing early convergence dynamics. Kuramoto (a) and EAPF consensus (b) converge rapidly; PCO variants (d,e) show extra flashes from the start.

5 Conclusion

EAPF consensus (event-based phase/frequency consensus with $g_p = 0.02$, $g_f = 0.02$) is selected as the **primary HIL candidate** for Stage 4B under the tested Stage 4A conditions. Kuramoto ($K = 5.0$) is selected as the **secondary comparison model**. EAPF tracker, PCO adaptive PRC, and PCO simple are retained as comparative references with identified limitations.

The primary HIL candidate achieves perfect (100%) strict zero-lag synchronisation across all 240 N=3 trials for that model (1200 trials total across all five models) and all 300 N=5 scalability trials, uses only flash-event-derived neighbour state (never true internal phases), and has stable performance across independent random seeds.

References